MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

WEEKLY CHAPTER TEST – 04 (SOLUTIONS)
Chapter 3: Pair of Linear Equations in Two Variables

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, every intermediate algebraic step, and boxed final answers with units.
2. In internal choice (**OR**) questions, both options are solved completely with identical CBSE step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Graph scales are maintained uniformly.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions
MATHS MANTHAN | (3 × 1 = 3 Marks)

Q1. The sum of the digits of a two-digit number is 9. If 27 is added to it, the digits get reversed. The number is: [1]

(A) 25 (B) 72 (C) 63 (D) 36

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]
Correct Answer: **(D) 36** [0.5 Mark]
Let tens digit $= x$, units digit $= y$: [0.5 Mark]
$x + y = 9$ and $(10x + y) + 27 = 10y + x \implies 9y - 9x = 27 \implies y - x = 3$. Adding both: $2y = 12 \implies y = 6, x = 3 \implies$ **36**.

Q2. Aruna has only Rs. 1 and Rs. 2 coins with her. If the total number of coins is 50 and the amount of money is Rs. 75, then the number of Rs. 1 and Rs. 2 coins are, respectively: [1]

(A) 35 and 15 (B) 35 and 20 (C) 15 and 35 (D) 25 and 25

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]
Correct Answer: **(D) 25 and 25** [0.5 Mark]
Justification: Let Rs. 1 coins $= x$, Rs. 2 coins $= y$. Then $x + y = 50$ and $1x + 2y = 75$. [0.5 Mark]
Subtracting first from second: $y = 25 \implies x = 50 - 25 = 25$. Hence **25** and **25**.

Q3. If $x = a, y = b$ is the solution of the equations $x - y = 2$ and $x + y = 4$, then the values of a and b are: [1]

(A) 3 and 1 (B) 3 and 5 (C) 5 and 3 (D) -1 and -3

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]
Correct Answer: **(A) 3 and 1** [0.5 Mark]
Justification: Substituting (a, b) gives $a - b = 2$ and $a + b = 4$. [0.5 Mark]
Adding: $2a = 6 \implies a = 3$. Then $3 + b = 4 \implies b = 1$. Hence **3** and **1**.

SECTION B: Assertion-Reason Question — MATHS MANTHAN | (1 × 1 = 1 Mark)

Q4. **Assertion (A):** The area of the rectangle formed by the lines representing $x = 8, y = 6$ with the coordinate axes is 48 sq. units.
Reason (R): The lines $x = 8$ and $y = 6$ intersect at $(8, 6)$, forming a right rectangle of length 8 units and breadth 6 units with the coordinate axes $x = 0$ and $y = 0$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** [0.5 Mark]

Verification: Vertices of rectangle bounded by $x = 0, x = 8, y = 0, y = 6$ are $(0, 0), (8, 0), (8, 6), (0, 6)$. [0.5 Mark]

Area $= 8 \times 6 = 48$ sq. units. Thus both (A) and (R) are true and (R) explains (A).

SECTION C: Very Short Answer Questions — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q5. Given $217x + 131y = 913$ and $131x + 217y = 827$, find the values of $(x + y)$ and $(x - y)$, and hence find the solution (x, y). [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $217x + 131y = 913$ — (i) and $131x + 217y = 827$ — (ii)

Step 1: Add-Subtract Trick for Symmetric Coefficients [1 Mark]

Adding (i) and (ii): $348x + 348y = 1740 \implies x + y = \frac{1740}{348} = 5$ — (iii)

Subtracting (ii) from (i): $86x - 86y = 86 \implies x - y = \frac{86}{86} = 1$ — (iv)

Step 2: Solving (iii) and (iv) [1 Mark]

Adding (iii) and (iv): $2x = 6 \implies x = 3$. Putting $x = 3$ in (iii): $y = 5 - 3 = 2$.

$$\boxed{\mathbf{x + y = 5}, \quad \mathbf{x - y = 1}, \quad \mathbf{(x, y) = (3, 2)}}$$

Q6. Determine whether the lines representing $2x - 3y = 8$ and $4x - 6y = 9$ are intersecting, coincident, or parallel. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $2x - 3y - 8 = 0$ and $4x - 6y - 9 = 0$. Here $a_1 = 2, b_1 = -3, c_1 = -8$ and $a_2 = 4, b_2 = -6, c_2 = -9$.

Formula / Theorem Used: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies$ Parallel Lines

Step 1: Comparing Ratios [1 Mark]

$$\frac{a_1}{a_2} = \frac{2}{4} = \frac{1}{2}, \qquad \frac{b_1}{b_2} = \frac{-3}{-6} = \frac{1}{2}, \qquad \frac{c_1}{c_2} = \frac{-8}{-9} = \frac{8}{9}$$

Step 2: Geometric Conclusion [1 Mark]

Since $\frac{1}{2} = \frac{1}{2} \neq \frac{8}{9}$, we get $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

$$\boxed{\textbf{The lines are parallel.}}$$

Q7. The taxi charges in a city consist of a fixed charge together with the charge for the distance covered. For a distance of 10 km, the charge paid is Rs. 105 and for a journey of 15 km, the charge paid is Rs. 155. What are the fixed charges and the charge per km? How much does a person have to pay for travelling a distance of 25 km? **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let fixed charge = Rs. x, charge per km = Rs. y.

Formula / Theorem Used: Total Fare = x + (Distance) × y

Step 1: Setting Up Linear Equations [1 Mark]

10 km trip: $x + 10y = 105$ — (i) and 15 km trip: $x + 15y = 155$ — (ii)

Step 2: Solving for x and y [1 Mark]

Subtracting (i) from (ii): $5y = 50 \implies y =$ Rs. 10/km. From (i): $x + 10(10) = 105 \implies x =$ Rs. 5.

Step 3: Total Charge for 25 km [1 Mark]

Fare $= 5 + 25(10) = 5 + 250 =$ Rs. 255.

Fixed = Rs. **5**, Per km = Rs. **10**, 25 km fare = Rs. **255**

Q8. A person invested some amount at 12% simple interest per annum and another amount at 10% simple interest per annum. He received a total yearly interest of Rs. 130. But if he had interchanged the amounts invested, he would have received Rs. 4 more as yearly interest. How much amount did he invest at each rate? **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Let amount invested at 12% be Rs. x and at 10% be Rs. y (for $T = 1$ year).

Formula / Theorem Used: Simple Interest (SI) $= \dfrac{P \times R \times T}{100}$

Step 1: Translating Interest Conditions into Equations [2 Marks]

Condition I: $\frac{12x}{100} + \frac{10y}{100} = 130 \implies 12x + 10y = 13000 \implies 6x + 5y = 6500$ — (i)

Condition II (interchanged): $\frac{10x}{100} + \frac{12y}{100} = 130 + 4 = 134 \implies 10x + 12y = 13400 \implies 5x + 6y = 6700$ — (ii)

Step 2: Solving via Add-Subtract Method [2 Marks]

Adding (i) and (ii): $11x + 11y = 13200 \implies x + y = 1200$ — (iii)

Subtracting (ii) from (i): $x - y = -200$ — (iv)

Step 3: Finding Both Investment Amounts [1 Mark]

Adding (iii) and (iv): $2x = 1000 \implies x = 500$. Substituting in (iii): $500 + y = 1200 \implies y = 700$.

Rs. **500** at **12%** and Rs. **700** at **10%**

Q9. CASE STUDY: Theme Park Ticketing Gate Analytics

On a public holiday, a popular amusement theme park offered special admission passes. The entry ticket for a child is priced at Rs. x and for an adult is priced at Rs. y. Exactly 100 children and 200 adults entered the park that day. The cashier collected a total sum of Rs. 60000. It is known that an adult ticket costs Rs. 150 more than a child ticket, i.e., $y - x = 150$. [4]

(i) Formulate the pair of linear equations in x and y representing the total collection and the price difference. [1]

(ii) Find the individual ticket price for a child (x) and for an adult (y). [1]

(iii) Verify the total number of visitors who entered the park. [1]

(iv) If the park offers a 20% festive discount on adult tickets next Sunday, find the revenue collected from 100 children and 200 adults. [1]

— OR —

Find the total revenue if 150 children and 150 adults visit at standard rates. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Formula / Theorem Used: Total Gate Revenue = (Child Count)x + (Adult Count)y

Sub-part (i): Formulating Linear Equations [1 Mark]

Total collection: $100x + 200y = 60000 \implies x + 2y = 600$ — (i) Price difference: $-x + y = 150$ — (ii)

Sub-part (ii): Finding Child (x) and Adult (y) Ticket Prices [1 Mark]

Adding (i) and (ii): $(x + 2y) + (-x + y) = 600 + 150 \implies 3y = 750 \implies y = \text{Rs. } 250$.

Substituting in (ii): $x = y - 150 = 250 - 150 = \text{Rs. } 100$. $\boxed{\mathbf{x = \text{Rs. } 100,\quad y = \text{Rs. } 250}}$

Sub-part (iii): Total Visitors [1 Mark]

Total Visitors = Children + Adults = $100 + 200 = \boxed{\mathbf{300}\text{ visitors}}$.

Sub-part (iv): Revenue with 20% Discount on Adult Tickets [1 Mark]

Discounted adult fare = $250 - \frac{20}{100}(250) = 250 - 50 = \text{Rs. } 200$.

Total Sunday revenue = $100(100) + 200(200) = 10000 + 40000 = \boxed{\text{Rs. } \mathbf{50{,}000}}$.

Sub-part (iv) [OR Option]: Revenue for 150 Children & 150 Adults [1 Mark]

Revenue = $150(100) + 150(250) = 15000 + 37500 = \boxed{\text{Rs. } \mathbf{52{,}500}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★